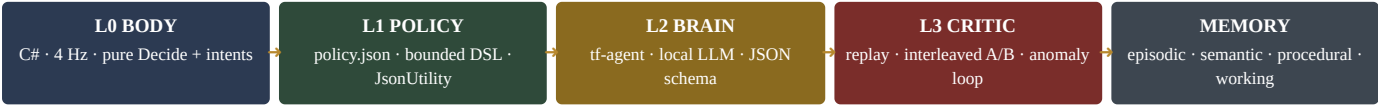


THRONEFALL AGENT

Design packet v3 (final) — Body · Policy · Brain · Critic — aware, remembering, self-correcting, legit



v3 incorporates all 10 items of V2-AUDIT.md plus 13 findings from three further review passes and live-log measurements from the “v1” install snapshot (AUDIT-LOG.md). Every identifier below exists in src/ or decompiled/; the identifier check, the cross-reference check and the arithmetic were re-run by script.

1. Ground truth from the code that shapes the design

Fact	Where	Consequence
wavenumber starts at -1 and increments in StartSpawning; by day the raw value is the last started night	EnemySpawner.cs:32, 553	Snapshot exposes NextWaveIndex; rules never use raw Wave
GetWaveInfoForNextWave() fills name, count, elite, maxHP, speed, range, damage, cooldown, gold, difficulty — the pause-menu preview. No flying field.	EnemySpawner.cs:600-745, WaveEnemyInfo.cs	P1 calls it on day/night edges; flying from prefab TaggedObject.Contains(ETag.Flying)
PresentChoices is called only by BuildSlot.ExecuteUpgrade; the exact upgrade sits in private upgradeSelected	BuildSlot.cs:148, 773-792	P2 scores build-branch choices exactly (reflection primary, Upgrades[Level] fallback); perk frames are a separate path
GetBuildClass sums income over all branches and ignores hpChange	BotPerception.cs:395-434	P2 scores the pending branch alone; P7 adds the Fortify class
Shrines charge only from real Hp deaths (any side, no knock-outs) inside collectionRange	Hp.cs:402-411, Shrine.cs:163-180	B3 = anchor bias (army.shrineBias), never a day walk
PotionVialAutoCast : ManualAttack; the bot pumps one handle only	Bot.cs:905-906	P6 pumps activeWeapon and passiveWeapon
HandleBlockingFrame returns before Decide() while a choice waits; DayTimeLeft keeps running	Bot.cs:281-293	No network on the choice path, ever
PerkManager.SetEquipped only edits the list; opening the loadout frame was the double-fire	PerkManager.cs:440-456, Bot.cs:356-390	A7 seeds via SetEquipped before the transition; respects maxPerkCount, IsUnlocked
net472 plugin; Unity ships JsonUtility; Newtonsoft not guaranteed	ThronefallTrainer.csproj	Policy/recorder use JsonUtility shapes (arrays, no dictionaries)
Snapshot holds Unity object refs (Coin, TaggedObject, ...)	BotPerception.cs:30-58	Split into SnapshotData (values, Vec2) + SnapshotRefs; tests need no game DLLs
“Tests — none exist”; 10 890 / 30 276 log lines are invalid	AUTOPILOT.md	Phase 0 = recorder + pure Decide + fixture replay; invalid → one scene-load note

1.3 Live-log evidence (Thronefall.zip “v1”: 37 355 lines, 17 process starts, ≈ 2.7 h, cfg = legit + autopilot on)

Measurement	Value	Consequence
invalid share	638 / 37 355 = 1.7 % (36 % was the older log)	D1 stays; the noise is now stuck:* (519 lines)
Nordfels stuck:3 + snap positions	136 of 150 in one 4 m cell at (0, 4), mode Sp endGold — the keep-interior slot	B2 keep anchor moves to Phase 1 (one anchors.json entry); first known target for add_nav_anchor
End-of-match	'After Match Frame ' (trailing space) closed by the generic branch 12/12; match-end never fired	BackToLevelSelectHelper lookup misses on this build; Phase 0 verifies; frames.json seeded: Level Up Frame, After Match Frame , Title Frame
army-placed · choice-pick · horn-interact	0 · 0 · 0	legit nights were fought by the hero alone; Phase 1 live check requires army-placed on Nordfels
teleport-nudge while cfg says legit	8 lines	undecidable today (older session or leak); recorder tags every line with legit
Nordfels progress · avg line	wave 12/13 reached, ≈ 24 min at 12/13 · 229 B @ 4 Hz ≈ 3.3 MB/h	final night dominates castleHpMin/durationNight; 2 Hz DTO carries 3× the fields for less volume

2. Architecture and invariants

Windows 11 · thronefall.exe · BepInEx 5

Capture() → SnapshotData + SnapshotRefs

Decide(in SnapshotData, in PolicyTable, ref BotMemory, now) → DecideResult

Tick executes DecideResult.Intents against SnapshotRefs

PolicyTable ← agent/policy.json JsonUtility · validated · last-good · 1 Hz mtime poll

Recorder → agent/runs/<runId>/{ticks.jsonl 2 Hz, events.jsonl, summary.json} · bounded queue · 1 writer thread

Mailbox → outbox.jsonl · alive.json ← inbox.jsonl (bounded commands) no sockets in the game

files, polled with byte offsets (no inotify over DrvFs)

WSL2 · Docker compose (or native `uv`)

tf-agent: poll → evidence pack → LLM (JSON schema) → validate → policy.json (temp + os.replace)

critic: replay harness · interleaved A/B evaluator · live detectors · action verifier

memory: agent/memory/{episodic,semantic,procedural,working} dashboard localhost:8787

LLM: local (LM Studio :1234 / Ollama :11434 / vLLM :8000) · MiniMax M3 (vision) · coder model (offline patches)

Invariant	Enforced by
Perception read-only	lint forbids InteractionBegin SetEquipped TeleportTo SetState in BotPerception.cs
Decide pure	no UnityEngine in BotBrain.cs/KnobTable.cs/PolicyTable.cs (lint); netstandard2.0 tests compile those files directly
Body never waits	no System.Net in the plugin (lint); last-good policy; recorder queue cap 2 000, drop-oldest, counted
Brain writes data, never code, into the live process	body reads four validated files only
Rules bounded · scoped · sourced · expirable · on trial	policy.schema.json; knob [min,max]; evaluator is sole writer of status
Legit gate first	knob allow-list has no cheat entries; intents tagged CheatOnly are refused when Legit

3. Body — Phase 0 seam and Phase 1 fields

// BotBrain.cs – pure, testable

enum IntentKind { BeginHold, ReleaseHold, PumpAttack, CommandArmy,

PlaceArmy, HornInteract, SwitchNight, TransitionLevel, SeedLoadout,

PickChoice, CloseFrame, ApplyFrame, BackToMap, MenuAdvance, None }

struct Intent { IntentKind Kind; int Index; bool CheatOnly; }

struct Vec2 { float X, Z; }

struct DecideResult { BotMode Mode; Vec2 AimPos; float Arrive;

Intent[] Intents; int[] RulesFired; }

static DecideResult Decide(in SnapshotData s, in PolicyTable p,

ref BotMemory m, float now);

// Tick (Bot.cs): Capture → Decide → execute intents via SnapshotRefs

// → watchdog/nav (AstarPath) → recorder

// Regression gate: fixtures recorded BEFORE the refactor must give the

// same mode sequence AFTER it.

ID	Adds to SnapshotData	Source
P1	NextWaveIndex, NW.Count/Elite Count/HpSum/MaxRange/MaxSpeed/HasFlying/HasRanged/GoldReward/Difficulty/SpawnLineCentroid	GetWaveInfoForNextWave(), GetNextWave(), prefab tag
P2	per pending choice: Pickable, IncomeDelta, HpDelta, Military, Cores	upgradeSelected.upgradeBranches[i].choiceDetails == choice
P3	Threat.Range/MinRange/IsRanged, same for Near	enemy AutoAttack.targetPriorities
P4	Shrines[] X,Z,Activated,Range(≤8)	Shrine at scene load
P5	PrefinalWave, LastWaveNext, CastleHpPct	EnemySpawner, castle Hp.HpPercentage
P6	HasPassiveWeapon, ActiveRange, ActiveFiresMoving	WeaponEquipper
P7	slotClass ∈ {Harvest, Military, Income, Fortify, Keep, Other}, HpDelta	UpgradeBranch.hpChange, objectsToActivate
P8	recorder: 394 B/line DTO · 2 Hz = 2.84 MB/h · caps 14 d / 40 runs / 300 MB	—

slotScore = ∑_k knob[k] · phase[k] · class_k(slot) + proximity(slot, NW.SpawnLineCentroid) · knob[score.choke]

choiceScore = choice.military·Military + choice.income·IncomeDelta + choice.hp·HpDelta (Pickable only; tie → first Pickable)

standoff = clamp(max(Threat.Range + stance.foeMargin, ActiveRange · stance.standoffFrac), 4, 14)

defaults reproduce today's numbers exactly (1000 / 100 / 30+3·min(Δ,10) / 10 / 0.7 / 0.5 / 11) → Phase 2 with no rules ≡ Phase 1

4. Policy file (JsonUtility-shaped) and validation

```
{ "schemaVersion": 1, "version": 17,
  "knobs": [
    { "name": "score.fortify", "def": 10, "min": 0, "max": 300 },
    { "name": "score.choke", "def": 0, "min": 0, "max": 150 },
    { "name": "choice.hp", "def": 40, "min": 0, "max": 400 },
    { "name": "stance.foeMargin", "def": 1.0, "min": 0, "max": 4 },
    { "name": "stance.retreatHp", "def": 0.5, "min": 0.3, "max": 0.7 },
    { "name": "army.depth", "def": 11, "min": 6, "max": 16 },
    { "name": "army.shrineBias", "def": 0, "min": 0, "max": 1 } ],
  "rules": [ {
    "id": "nordfels-walls-late", "scene": "Nordfels",
    "waveMin": 2, "waveMax": 99, "phase": "day",
    "when": [ { "field": "NW.Count", "op": "ge", "value": 20 },
              { "field": "Fortify.Count", "op": "lt", "value": 2 } ],
    "then": [ { "knob": "score.fortify", "op": "add", "value": 120 } ],
    "status": "trial", "confidence": 0.6, "createdBy": "reflection",
    "created": "2026-09-29", "expires": "2026-11-01",
    "evidence": [ "20260929T211400Z-Nordfels" ] } ] }
```

Validator (body, ~200 lines; any failure → last-good + policy-reject)

- 1 · schemaVersion==1; version > loaded (rollback = new higher version with old content)
- 2 · every knob name in the compiled allow-list; min ≤ def ≤ max
- 3 · every when.field in generated SNAPSHOT-FIELDS.md / FieldIndex.cs (switch, not reflection)
- 4 · op ∈ {lt,le,eq,ge,gt,in}; then.op ∈ {add,set,mul}; phase ∈ {day,night,any}
- 5 · ≤ 40 live rules per scene; ≤ 3 live rules per knob per overlapping scope
- 6 · expires in the future; status ∈ {trial,promoted,retired}
- 7 · results clamped to [min,max] at evaluation, always

Load: sidecar temp + os.replace; body FileShare.ReadWrite|Delete, 1 Hz mtime poll, parse off-thread, swap on main; C# and Python validators agree on 20 shared fixtures

5. Memory and prompt budget

Store	Path	Content · retention
Episodic	runs/<runId>/ · memory/episodic/index.jsonl	ticks 2 Hz (14 d / 40 runs / 300 MB), events, summary.json; index kept forever
Semantic	policy.json · lessons.jsonl · scenes/<scene>.md	executable rules · reflections tagged confirmed/refuted/unknown · generated playbooks
Procedural	memory/procedural/skills/<name>/	patch.diff + test + provenance · merged by PR only
Working	core.md ≤ 1 200 tok · context.json	rebuilt per call: last 8 events, current summary, active trial

Reflection prompt: 1 200 + 3 500 + 1 800 + 900 + 600 = **8 000 tokens in**, ≤ 700 out. Est. on a 32B Q4 model, RTX 3090 Ti: 8 000/900 + 700/22 ≈ **41 s** of a 60 s budget. Ticks are never in the prompt; the brain calls query_ticks when it needs detail.

6. Brain — events, closed tool set, providers

Event	Brain task	Budget → fallback
run-start	pick_loadout from index + playbook; ≤ maxPerkCount, IsUnlocked; pinned while a trial runs on that scene	5 s → today's seeding
day-start	≤ 3 ephemeral knobs for this run	10 s → none
match-end	reflection: ≤ 3 lessons, ≤ 2 proposed rules, ≤ 1 retirement request	60 s, offline
anomaly	one action from the menu (§7)	20 s → flag_for_human
unknown-frame	request_screenshot → classify → close/apply/back + frames.json	10 s → generic close
policy-reject	repair or roll back (new higher version)	—

tools

propose_rule · retire_rule · set_ephemeral_knob · write_lesson

tools

pick_loadout · request_level · add_nav_anchor · park_slot

tools

query_ticks · query_runs · request_screenshot

tools

flag_for_human — closed set, lint-enforced

```
# config/agent.yaml - requirement: OpenAI-compatible + JSON-schema output; model names are env vars, probed by doctor.ps1
provider: local
local: { baseUrl: http://host.docker.internal:1234/v1, model: ${LOCAL_INSTRUCT_MODEL}, vision: false }
minimax: { baseUrl: https://api.minimax.io/v1, model: MiniMax-M3, apiKeyEnv: MINIMAX_API_KEY, vision: true }
coder: { baseUrl: http://host.docker.internal:1234/v1, model: ${LOCAL_CODER_MODEL} } # G4 offline patches only
routing: { reflection: local, anomaly: local, day-start: local, unknown-frame: minimax, skill-patch: coder }
timeouts: { reflection: 60, anomaly: 20, unknown-frame: 10, day-start: 10, run-start: 5 }
```

7. Critic — replay, interleaved A/B, anomaly loop

```
# evaluator — per scene, one trial at a time, pinned loadout
schedule  B A B A B A      (B baseline, A baseline + trial rule)
pairs      min 3, max 6
primary    win (0/1), then castleHpMin (0..1)
secondary  heroDeaths, goldUnspentAtDuskAvg, stalls, durationNight,
promote     Σ(win_A - win_B) ≥ 0
            AND mean(castleHpMin_A - castleHpMin_B) ≥ -0.05
            AND ≥ 1 secondary improves in ≥ 2 pairs
retire      Σ(win_A - win_B) ≤ -2 (early stop)  OR  6 pairs, no promotion
honesty     6 pairs cannot see a 10 % win-rate change; the bar is
            "not worse, and moved a secondary". thresholds = config.
```



Replay harness (Phase 0): xunit on netstandard2.0, compiles BotBrain.cs, KnobTable.cs, PolicyTable.cs only. Fixtures: neuland-tutorial, nordfels-w0-3, craaghelm-core-stall, recorded live before the refactor. Asserts: identical mode sequence pre/post; day drains to bld==0 before SwitchNight; no CheatOnly intent under Legit; rule firing under a given policy. A proposed rule that flips a slot pick on a **winning** fixture is marked *risky*.

Signature	Detector (1 s poll over outbox)	Allowed actions · verifyWithin
day never ends	bld > 0 > 240 s, no build-stall/switch-night	park_slot · nav.holdSeconds=4 · 60 s
nav wedge	stuck:3 + snap within 8 m > 30 s	add_nav_anchor · next visit
UI churn	> 6 frame-close / 20 s on one frame	request_screenshot → frames.json · 30 s
wave grind	same scene, 3 defeats, no rule change	request_level · retire_rule · pick_loadout · 1 run
legit leak	teleport-nudge while legit=true	flag_for_human only
policy churn	> 3 policy-loaded / 60 s	pause brain writes 10 min · flag_for_human
sidecar stale	alive.json older than 30 s	body keeps last-good · dashboard red

Each action records expectedEffect + verifyWithin; a failed action is never retried on the same signature in the same run; the next step is flag_for_human.

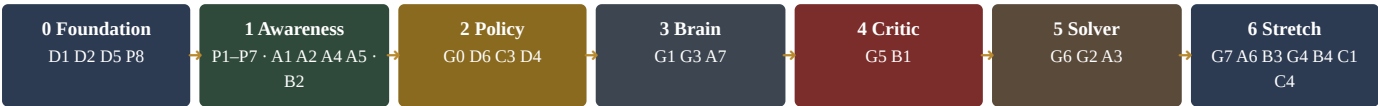
8. Transport, camera, security

Topic	Decision
Files across WSL2	Body writes under BepInEx/plugins/agent/ with FileShare.ReadWrite Delete; sidecar polls (watchfiles force_polling, 1 s, byte offsets), never renames a file the body appends to; writes via temp + os.replace; run-dev.ps1 -Native runs the sidecar as a Windows process without Docker
Camera	L0 Snapshot is truth · L1 ScreenCapture.CaptureScreenshotAsTexture() (module ref in csproj) only on unknown-frame and the wipe frame, ≤ 768 px, deleted after use · L2 synthetic 256 px minimap rendered from ticks · never streaming or pixels on the decision path
Security	no sockets or downloaded code in the game; sidecar mounts agent/ only, no ThroneSave.sav; keys in .env (doctor refuses keys in YAML); screenshots gitignored; multiplayer guard (C5); closed tool set

9. Repository, scripts, gates

	Script	Fails when
src/ThronefallTrainer/	Plugin Patches Bot BotPerception BotPatches BotBrain(pure) KnobTable PolicyTable Recorder Mailbox WaveIntel ChoiceIntel FieldIndex(gen) mailbox evidence providers brain evaluator detectors memory dashboard · pyproject.toml	
src/tf-agent/	xunit netstandard2.0	
tests/Replay.Tests/	3 real recordings	
tests/fixtures/runs/	pytest · integration/ vs real local endpoint	
tests/agent/	docs/ scripts/ docker/ config/ tools/ env/.env.example	
docs/	CI: lint.ps1 + build.ps1 + test.ps1 on Windows + Ubuntu runners. Plugin builds in CI against a csc -refonly reference assembly made from decompiled/*.cs (Roslyn body-strip); fallback = self-hosted Windows runner with the game installed. Never shipped.	
	doctor.ps1	.NET SDK < 8 · GameDir lacks exe / Harmony / JsonSerializerModule · no Python 3.12 · no Docker and no uv · port 8787 busy · a provider fails the schema probe · key found in YAML
	run-dev.ps 1	build fails · deploy fails · dashboard /health not green in 60 s
	test.ps1	replay red · pytest red · integration red (llm profile up) · smoke: fixture summary → validated policy.json not within 90 s
	lint.ps1	bot-lint FAIL (legit gate, widths, unscaled time, refs, enum, fields, purity, knob allow-list, no System.Net) · ruff · mypy --strict · PSScriptAnalyzer · forbidden patterns · gen-fields drift — any warning
	build.ps1 / release.ps 1	TreatWarningsAsErrors · zip + SHA256SUMS · sidecar must boot from the zip on a clean WSL distro and pass smoke

10. Phases — Definition of Done



Phase	Done when (all hold)
0	pre/post-refactor mode sequences identical on all 3 fixtures · bot-lint 0 FAIL · <code>doctor.ps1</code> PASS on a clean clone · <code>test.ps1</code> green · CHANGELOG entry
1	replay: Nordfels day 2 top slot is <i>Fortify</i> when <code>NW.Count ≥ 20</code> · choices branch-scored (fallback on ties only) · kite uses <code>Threat.Range</code> · both weapons pumped · one live Nordfels run with new fields non-zero, <code>army-placed</code> present, <code>snap</code> at (0, 4) ≤ 5 (was 64)
2	empty rule list ⇒ scores identical to Phase 1 · hand-written rule changes live scoring < 1 s · malformed file ⇒ last-good + <code>policy-reject</code> · C#/Python validators agree on 20 fixtures
3	smoke on local model · 3 consecutive matches ⇒ ≤ 2 schema-valid trial rules each · dashboard shows runs, rules, alive
4	injected bad rule retired within ≤ 6 pairs unattended · injected good rule promoted · <code>scenes/Nordfels.md</code> regenerates from data
5	wedge fixture ⇒ <code>add_nav_anchor</code> ⇒ clears in replay · unknown frame classified once via screenshot · legit leak always flags human

Phases 0–2 are pure C# and are the empirical gate: if hand-written rules in Phase 2 do not change a Nordfels day buy, no LLM will either.

11. Failure modes

Component	Failure	Detection	Recovery
Recorder	disk / IO error · queue overflow	exception counter · <code>dropped>0</code>	stop recording, keep playing, one event · drop-oldest, reported in summary
Policy	invalid · stale version	validator · version check	last-good + <code>policy-reject</code> · ignored
Mailbox	file locked	<code>IOException</code>	retry next second; never block Tick
Sidecar	down	<code>alive.json > 30 s</code>	body unaffected; detectors resume from byte offset
Provider	timeout · bad JSON	schema validation	per-event fallback; 3 failures ⇒ degraded 10 min
Evaluator	confounded pair (loadout changed)	pin check	pair discarded
Game patch	API drift	<code>decompile.ps1</code> diff + bot-lint	unresolved fields logged once, read as 0; rules on them never fire

Open items only running code settles: *csc* - refonly over this build's decompile · real local-model speed (≈ 19 s slack) · whether `FindObjectOfType<Shrine>` at load sees late-spawned shrines (else re-scan at day edge) · whether the 8 *teleport-nudge* lines were a legit leak (recorder *legit* tag decides next run).

12. Crosswalk — v1 idea → v3 fate (evidence)

v1	v3	Evidence
A1 name rubric for choices	P2 exact branch scoring + <code>choice.*</code> knobs	<code>BuildSlot.cs:782</code> , <code>upgradeSelected</code>
A2 wave-index scoring	P1 composition + P7 Fortify + <code>score.choke</code>	<code>EnemySpawner.cs:600</code> ; <code>GetBuildClass</code> ignores <code>hpC</code> hange
A5 potion system	pump both <code>ManualAttack</code> slots	<code>PotionVialAutoCast</code> , <code>Bot.cs:905</code>
B3 walk to shrine	kill-zone anchor bias	<code>Shrine.DeathOfUnitAt</code> via <code>Hp.cs:410</code>
B1 markdown playbooks	generated <code>scenes/*.md</code> from <code>policy.json</code> + lessons	free text was unvalidatable
D5 mock harness, week 3	pure <code>Decide</code> + fixture replay, Phase 0	“Tests — none exist”
G1 free-text lessons, MiniMax on the choice path	DSL + trial rules + interleaved evaluator; no network on any UI path	Reflexion needs an evaluator; <code>DayTimeLeft</code> runs during choices
G3 VPS dashboard	Docker on WSL2 or native, polled files	local-first; <code>DrvFs</code> inotify
new	A7 legit loadout · B2 keep anchor (Phase 1) · G5 evaluator · G6 anomaly loop · <code>NextWaveIndex</code> · <code>CastleHpPct</code>	live log: 136/150 wedges at one keep cell; 0 <code>army-placed</code>

Verification performed on this packet (details in `AUDIT-LOG.md §3`): every backticked identifier *grep-checked* against *src/* and *decompiled/*; phase membership cross-checked by script between the design and the backlog; recorder, prompt and latency arithmetic recomputed; every PDF page inspected.